Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

Reviewed example = a checked experiment; GenAge annotation = catalogue evidence only.

All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C001 **AMPK** 3 genes / 3 species 2 direct pairs	**Y1 SNF1** GenAge 619 **Reviewed example** Overexpression **Replicative life ↓** PMID 10921902	**W1 aak-2** GenAge 1 **Reviewed example** Extra gene copies **Mean life ↑** PMID 15574588	**F1 SNF1A** GenAge 1846 **Reviewed example** Adult tissue expression **Adult life ↑** PMID 23312286	No gene in this candidate network

Direct pairs: W1–F1 (1:1); F1–Y1 (1:1)

Yeast result is replicative lifespan; worm/fly interventions differ. No direct yeast–worm pair is implied.

C004 **ATG7** 3 genes / 3 species 3 direct pairs	**Y1 ATG7** GenAge 1218 **Reviewed example** Deletion **Chronological life ↓** PMID 19302372	**W1 atg-7** GenAge 23 **Reviewed example** RNAi in eat-2 **DR benefit reduced‡** PMID 17912023	**F1 Atg7** GenAge 21 **Reviewed example** Null mutation **Adult life ↓** PMID 18056421	No gene in this candidate network

Direct pairs: W1–F1 (1:1); W1–Y1 (1:1); F1–Y1 (1:1)

‡ Worm cell: dietary-restricted eat-2 mutants, not a general wild-type result. GenAge labels differ from this selected experiment; timing and background matter.

C080 **SOD2** 4 genes / 4 species 6 direct pairs	**Y1 SOD2** GenAge 623 **Reviewed example** Deletion in sch9Δ **CLS benefit lost§** PMID 12586694	**W1 sod-2** GenAge 985 **Reviewed example** Deletion **Adult life ↑** PMID 19197346	**F1 Sod2** GenAge 624 **Reviewed example** RNAi **Adult life ↓** PMID 12456885	**M1 Sod2** GenAge 827 **Reviewed example** Connective-tissue KO **Survival ↓** PMID 21108731

Direct pairs: W1–F1 (m:1); W1–M1 (m:1); W1–Y1 (n:1); F1–M1 (1:1); F1–Y1 (1:1); M1–Y1 (1:1)

§ Yeast: loss of sch9Δ chronological lifespan extension. Mouse: connective-tissue deficiency. Worm deletion effects depend on genetic background.

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.

Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.

GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.

Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.